Mapping AI Exposure to Norwegian Occupations

Construction of four exposure measures on STYRK-08

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Roadmap

- **Eloundou et al. (2024):** task-level LLM exposure rated by humans and GPT-4
- **Handa et al. (2025):** revealed usage from Claude.ai conversations
- **Felten et al. (2021):** AI progress on the abilities required by each occupation
- **Anthropic (2026):** time-weighted observed Claude usage at occupation level

Common destination: a 4-digit STYRK-08 score for every Norwegian occupation.

Why a mapping step is needed

- All four measures are built on US occupation codes (SOC).
- Norwegian register data use STYRK-08.
- STYRK-08 $\equiv$ ISCO-08 at 4 digits (SSB Notater 17/2011).
- Bridge: SOC $\rightarrow$ ISCO-08 via BLS crosswalk (Aug 2012, rev. June 2015).

O*NET: the US occupation database underneath three of the four measures

- Maintained by the US Department of Labor; covers ∼1,000 occupations on SOC 2018 codes (O*NET-SOC 2019 taxonomy)
- Each occupation has structured content:
  - **Tasks:** discrete activities (∼ 19,000 across all occupations; Pharmacists: 21 tasks, e.g. "review prescriptions for accuracy"). Tagged *Core* or *Supplemental*; no numerical weight.
  - **Abilities:** 52 underlying attributes (e.g. Deductive Reasoning, Manual Dexterity). Scored on prevalence and importance per occupation.
  - Also: skills, knowledge, work activities, work context (not used by the four measures here).
- Eloundou et al. and Handa operate on **tasks**; Felten operates on **abilities**; Anthropic 2026 operates at the occupation level.

Eloundou et al. (2024)

What Eloundou et al. measure

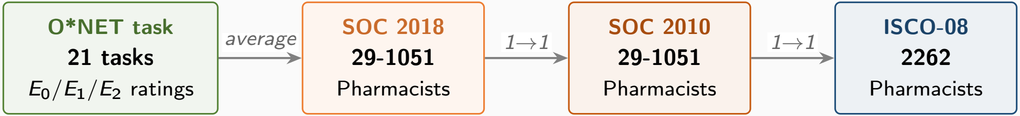
- Each O*NET task rated by humans and GPT-4 on three categories:
  - E_0 no exposure
  - E_1 LLM alone reduces task time by $\geq 50\%$
  - E_2 LLM with complementary software reduces task time by $\geq 50\%$
- Task-level score: $\beta_{\text{task}} = 0$ if E_0, 1 if E_1, 0.5 if E_2
- Occupation score: $\beta_{\text{occ}} =$ mean of β_{task} with O*NET Core tasks weighted 2, Supplemental tasks 1
- We use the published occupation-level file (`occ_level.csv`); it differs slightly from unweighted task means where an occupation has Supplemental tasks

The code chain

$$
\begin{tikzcd}
\begin{array}{c}\textbf{O*NET task}\\ \textbf{8823, \ldots}\\ \text{rated } E_0/E_1/E_2\end{array} \arrow[r, "average"] & \begin{array}{c}\textbf{SOC 2018}\\ \textbf{29-1051}\\ \text{Pharmacists}\end{array} \arrow[r, "BLS\ 2017"] & \begin{array}{c}\textbf{SOC 2010}\\ \textbf{29-1051}\\ \text{Pharmacists}\end{array} \arrow[r, "BLS\ 2012"] & \begin{array}{c}\textbf{ISCO-08}\\ \textbf{2262}\\ \text{Pharmacists}\end{array}
\end{tikzcd}
$$

Overlapping 4-digit unit groups in STYRK-08 are code-compatible with ISCO-08 (SSB Notater 17/2011). STYRK 2262 is Farmasøyter.

Worked example: Pharmacists (one-to-one mapping)



- Pharmacists carry the same 6-digit SOC in both vintages: 29-1051
- BLS crosswalk sends 29-1051 to a single ISCO-08 code (2262)
- Norwegian STYRK 2262 is Farmasøyter, no Norway-specific re-coding

Task-level scoring for Pharmacists (SOC 29-1051, all 21 tasks)

O*NET task	E	β
Review prescriptions for accuracy	E_2	0.5
Assess identity, strength, purity of meds	E_0	0.0
Provide info on drug interactions	E_2	0.5
Analyze prescribing trends	E_2	0.5
Maintain pharmacy files and patient profiles	E_2	0.5
Collaborate with other health professionals	E_2	0.5
Plan procedures for mixing and labeling	E_0	0.0
Order and purchase pharmaceutical supplies	E_2	0.5
Compound and dispense medications	E_0	0.0
Contact insurance companies on billing	E_2	0.5
Advise customers on medication brands	E_2	0.5
Teach pharmacy students serving as interns	E_1	1.0
Provide services for diabetes, asthma, etc.	E_2	0.5

O*NET task	E	β
Refer patients to other professionals	E_2	0.5
Work in hospitals/clinics or for HMOs	E_2	0.5
Update or troubleshoot pharmacy databases	E_1	1.0
Manage pharmacy operations, hire, supervise	E_2	0.5
Prepare sterile solutions	E_0	0.0
Offer health promotion / prevention activities	E_0	0.0
Assay radiopharmaceuticals	E_0	0.0
Publish educational information	E_1	1.0

Counts: 6 E_0 3 E_1 12 E_2

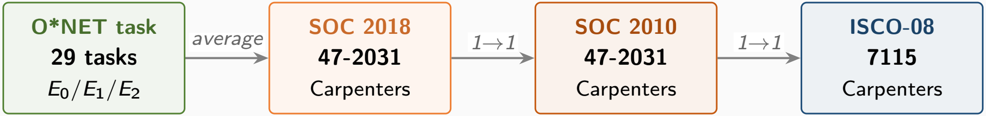
Occupation β (mean of 21): 0.4286 (Q4)

Other one-to-one mappings (large Norwegian occupations)

STYRK	Norwegian title	SOC 2010 (US title)	employed	β_{occ}	Q
7115	Tømrere og snekkere	47-2031 Carpenters	47,106	0.144	2
2330	Lektorer (videregående skole)	25-2031 Secondary teachers	31,155	0.333	3
2611	Jurister og advokater	23-1011 Lawyers	11,808	0.425	4
2631	Forskere/rådgivere, samf.økonomi	19-3011 Economists	1,282	0.553	5
3511	Driftsteknikere, IKT	43-9011 Computer Operators	16,517	0.736	5

All five are clean one-to-one in the BLS crosswalk. Exposure spans Q2 to Q5: carpenters at the bottom, ICT operations technicians at the top. β_{occ} is taken from the published Core-weighted occupation file, so it can differ slightly from an unweighted task mean. SOC titles do not always match the Norwegian title literally (e.g. "Computer Operators" vs. "Driftsteknikere, IKT").

Worked example: Tømrere og snekkere (STYRK 7115)



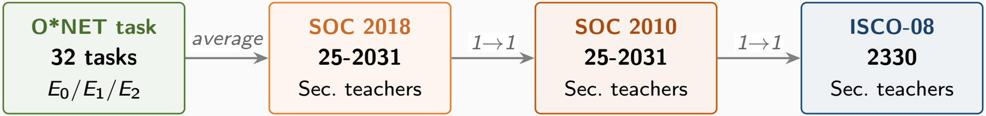
STYRK 7115 is Tømrere og snekkere. Same 6-digit SOC in both vintages, one O*NET specialty, 29 tasks. Mostly E_0: physical work dominates.

Task-level scoring: Carpenters (SOC 47-2031, 29 tasks)

O*NET task	E	β	O*NET task	E	β
Follow safety rules and regulations	E_0	0.0	Dig or direct digging of post holes	E_0	0.0
Measure and mark cutting lines on materials	E_0	0.0	Cover subfloors with building paper	E_0	0.0
Assemble and fasten materials	E_0	0.0	Construct forms or chutes for concrete	E_0	0.0
Study specifications in blueprints	E_2	0.5	Arrange for subcontractors	E_2	0.5
Shape or cut materials to measurements	E_0	0.0	Build or repair cabinets, doors, floors	E_0	0.0
Verify trueness with plumb bob and level	E_0	0.0	Finish surfaces of woodwork or wallboard	E_0	0.0
Inspect ceiling/floor tile, wall coverings	E_2	0.5	Select and order lumber and materials	E_2	0.5
Erect scaffolding or ladders	E_0	0.0	Work with or remove hazardous material	E_0	0.0
Install windows, frames, fixtures	E_0	0.0	Fill cracks or defects in plaster	E_0	0.0
Maintain records, present written reports	E_1	1.0	Prepare cost estimates for clients	E_2	0.5
Remove damaged or defective parts	E_0	0.0	Perform minor plumbing or concrete work	E_0	0.0
Maintain job records, schedule work crew	E_2	0.5	Apply shock-absorbing or sound-deadening	E_0	0.0
Anchor and brace forms and structures	E_0	0.0	Examine structural timbers for decay	E_0	0.0
Bore boltholes in timber/masonry/concrete	E_0	0.0	Build sleds from logs and timbers	E_0	0.0
Install rough door/window frames, subflooring	E_0	0.0			

Counts: 22 E_0 1 E_1 6 E_2 SOC β (mean of 29 tasks): 0.138 (Q2)

Worked example: Lektorer videregående (STYRK 2330)



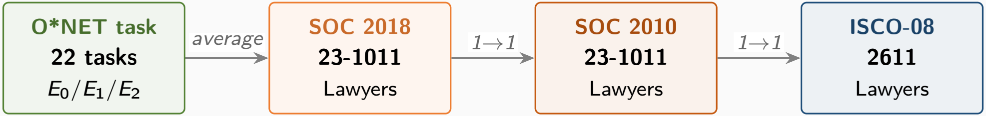
STYRK 2330 is Lektorer mv. (videregående skole). One O*NET specialty, 32 tasks. Most are E_2 (LLM with software is enough) and reflect classroom and administrative work.

Task-level scoring: Secondary teachers (SOC 25-2031, 32 tasks)

O*NET task	E	β	O*NET task	E	β
Instruct through lectures, discussions	E_2	0.5	Plan and conduct field trips	E_0	0.0
Establish rules of student behavior	E_0	0.0	Sponsor extracurricular activities	E_0	0.0
Prepare materials and classrooms	E_2	0.5	Perform administrative duties	E_2	0.5
Adapt teaching to varying abilities	E_2	0.5	Prepare and implement remedial programs	E_2	0.5
Observe and evaluate students' performance	E_2	0.5	Attend professional meetings and workshops	E_0	0.0
Establish clear objectives for lessons	E_2	0.5	Use computers and audiovisual aids	E_2	0.5
Prepare students for later grades	E_2	0.5	Select, store, order, or use class materials	E_2	0.5
Prepare, administer, and grade tests	E_2	0.5	Collaborate with other teachers	E_2	0.5
Maintain accurate, complete records	E_2	0.5	Plan and supervise projects, field studies	E_2	0.5
Enforce attendance and progression rules	E_0	0.0	Meet with professionals on programs	E_0	0.0
Instruct and monitor use of equipment	E_0	0.0	Guide and counsel students	E_2	0.5
Prepare content for diverse-needs students	E_2	0.5	Confer with parents on progress	E_0	0.0
Assign and grade class work and homework	E_2	0.5	Meet with administrators on instruction	E_2	0.5
Prepare reports on student progress	E_2	0.5	Prepare for assigned classes	E_2	0.5
Keep informed about developments in education	E_1	1.0	Provide disabled students with devices	E_2	0.5
Prepare objectives and outlines	E_2	0.5	Administer standardized tests	E_0	0.0

Counts: 12 E_0 1 E_1 19 E_2 **SOC β (mean of 32 tasks): 0.328 (Q3)**

Worked example: Jurister og advokater (STYRK 2611)



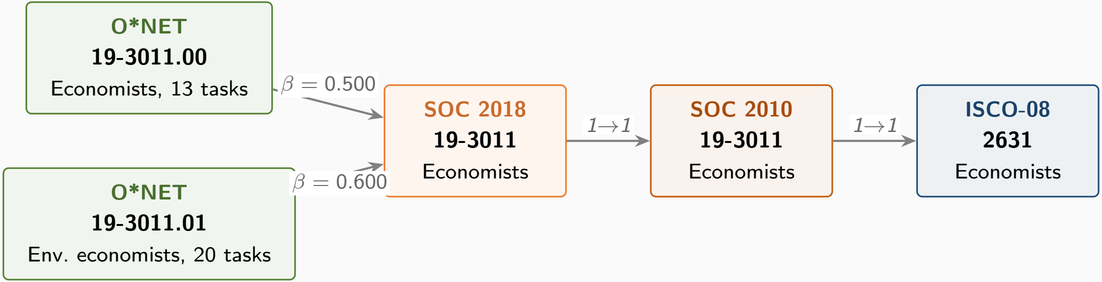
STYRK 2611 is Jurister og advokater. One O*NET specialty, 22 tasks. Nearly all tasks are E_2: text-heavy work where LLM plus software can substitute.

Task-level scoring: Lawyers (SOC 23-1011, 22 tasks)

O*NET task	E	β	O*NET task	E	β
Advise clients on legal rights	E_2	0.5	Confer with colleagues with specialties	E_0	0.0
Represent clients in court or in hearings	E_0	0.0	Examine legal data for advisability	E_2	0.5
Select jurors, argue motions, meet judges	E_0	0.0	Search and examine public records	E_2	0.5
Study Constitution, statutes, decisions	E_2	0.5	Act as trustee, guardian, or executor	E_2	0.5
Prepare legal briefs and opinions	E_2	0.5	Help establish company legal policies	E_2	0.5
Interpret laws, rulings, regulations	E_2	0.5	Negotiate settlements of civil disputes	E_2	0.5
Prepare and draft legal documents	E_2	0.5	Probate wills, represent estate executors	E_2	0.5
Present and summarize cases to judges	E_2	0.5	Work in environmental, public health law	E_2	0.5
Gather evidence by interviewing clients	E_2	0.5	Perform administrative and management	E_2	0.5
Analyze the probable outcomes of cases	E_2	0.5	Supervise legal assistants	E_2	0.5
Evaluate findings; develop defense strategy	E_2	0.5	Provide pro bono services	E_2	0.5

Counts: 3 E_0 0 E_1 19 E_2 SOC β (mean of 22 tasks): 0.432 (Q4)

Worked example: Forskere/rådgivere, samf.økonomi (STYRK 2631)



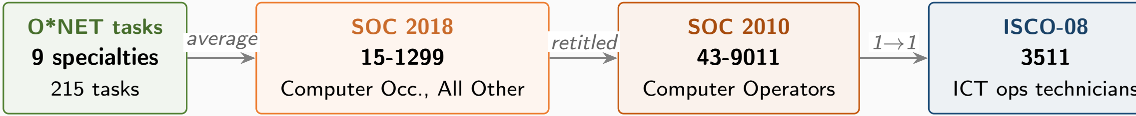
Two O*NET specialties roll up to one SOC 2018 code. The SOC-level β averages the two specialty means: $(0.500 + 0.600)/2 = 0.550$, regardless of how many tasks each specialty contains.

Task-level scoring: Economists (SOC 19-3011, 33 tasks across 2 specialties)

19-3011.00 Economists (13 tasks)			**19-3011.01 Env. Economists (20 tasks)**		
Study economic and statistical data	E_2	0.5	Prepare and deliver presentations	E_2	0.5
Conduct research on economic issues	E_2	0.5	Develop programs or policy recommendations	E_2	0.5
Compile, analyze, report data	E_2	0.5	Develop economic models and forecasts	E_2	0.5
Supervise research projects	E_2	0.5	Demonstrate economic benefits of policies	E_2	0.5
Teach theories and methods of economics	E_2	0.5	Conduct research on environmental relations	E_2	0.5
Study impacts of public policies	E_2	0.5	Perform complex mathematical analyses	E_1	1.0
Formulate recommendations and plans	E_2	0.5	Write social, legal, economic impact reports	E_1	1.0
Explain economic impact of policies	E_2	0.5	Teach courses in environmental economics	E_2	0.5
Provide advice and consultation	E_2	0.5	Develop programs to promote sustainability	E_2	0.5
Forecast production and consumption	E_2	0.5	Develop systems for analyzing data	E_2	0.5
Develop economic guidelines and standards	E_2	0.5	Write research proposals and grants	E_1	1.0
Testify at regulatory or legislative hearings	E_2	0.5	Examine exhaustibility of natural resources	E_2	0.5
Provide litigation support	E_2	0.5	Monitor or analyze market and env. trends	E_2	0.5
Write technical documents or articles	E_1	1.0	Develop environmental research project plans	E_2	0.5
			Conduct research on economic, env. topics	E_2	0.5
			Collect and analyze data on env. impact	E_2	0.5
			Assess costs and benefits of activities	E_2	0.5
			Identify environmentally friendly business	E_2	0.5
			Interpret indicators of ecosystem health	E_2	0.5
			Collaborate with researchers across fields	E_2	0.5
Specialty mean: 0.500			**Specialty mean:** 0.600		

SOC $\beta = (0.500 + 0.600)/2 =$ **0.550 (Q5)**

Worked example: Driftsteknikere, IKT (STYRK 3511)

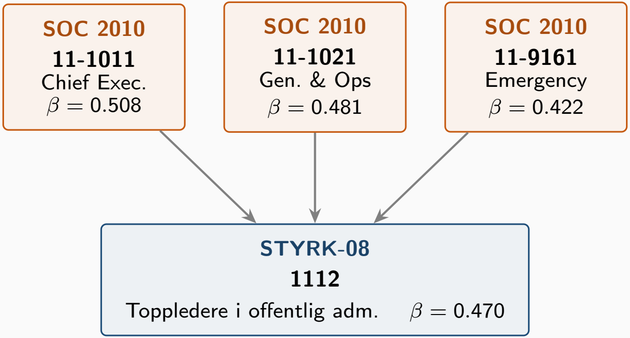


The SOC 2010 → 2018 re-coding turned "Computer Operators" into the catch-all "Computer Occupations, All Other" (15-1299), which contains nine sub-specialties. The Eloundou et al. pipeline averages those nine specialty means; tasks are not aggregated directly.

Specialty-level scoring: Computer Occupations, All Other (15-1299)

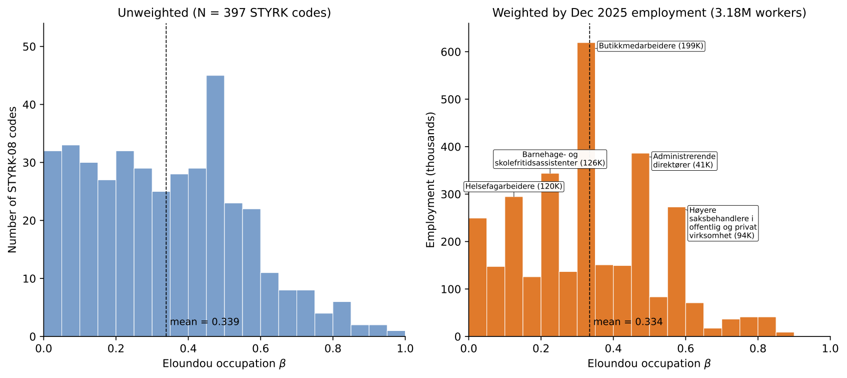
O*NET-SOC	Title	N tasks	specialty β
15-1299.01	Web Administrators	23	0.913
15-1299.02	Geographic Information Systems Technologists	27	0.593
15-1299.03	Document Management Specialists	18	0.611
15-1299.04	Penetration Testers	22	0.795
15-1299.05	Information Security Engineers	20	0.675
15-1299.06	Digital Forensics Analysts	20	0.725
15-1299.07	Blockchain Engineers	17	0.971
15-1299.08	Computer Systems Engineers/Architects	28	0.768
15-1299.09	Information Technology Project Managers	21	0.571
SOC β (mean of 9 specialties): 0.736 (Q5)			

Worked example: many-to-one mapping



- Three SOC 2010 codes map to ISCO 1112 in the BLS crosswalk
- STYRK β is the unweighted mean of contributing SOC-level β
- 57.7% of Eloundou et al. STYRK codes have at least one contributor flagged as a partial match in the BLS file

Distribution of Eloundou et al. β across Norwegian occupations



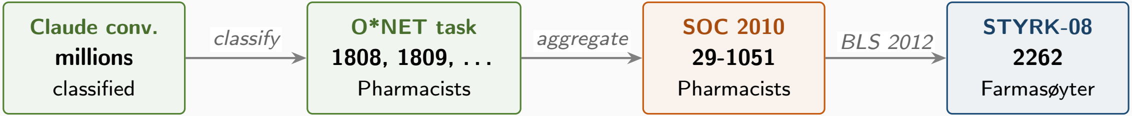
Left: one observation per STYRK-08 code (397 codes; mean 0.34). Right: the same distribution weighted by December 2025 employment (3.18M workers; mean 0.33); the largest single occupations are labelled. Large occupations sit across the whole exposure range, from health care workers (0.11) to policy administration professionals (0.57).

Handa et al. (2025)

What Handa et al. measure

- Sample of Claude.ai conversations classified into O*NET tasks
- Occupation exposure = share of conversations matched to that occupation's tasks
- Each conversation also classified as:
  - **automation:** directive, user issues a request, accepts the output
  - **augmentation:** iterative feedback, refinement
- Released as the Anthropic Economic Index (April 2025)

The code chain



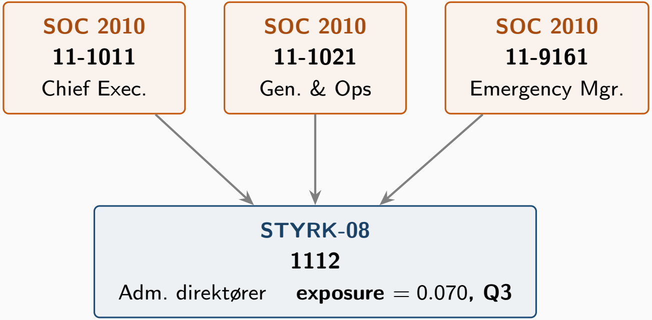
No SOC 2018 step: the Anthropic Economic Index reports occupation labels on SOC 2010.

Worked example: Pharmacists (SOC 29-1051)

O*NET task	share	directive	feedback + iter.
Provide info on drug interactions and side effects	0.0762	0.081	0.036
Assess identity, strength, or purity of medications	0.0412	0.112	0.060
Compound and dispense medications	0.0123	0.167	0.000
Publish educational information for pharmacists or patients	0.0091	0.000	0.000
Plan procedures for mixing, packaging, or labeling pharmaceuticals	0.0041	0.000	0.000
Provide services for patient management of conditions	0.0036	0.000	0.000
Analyze prescribing trends to monitor patient compliance	0.0016	0.000	0.000
. . . *13 of 20 tasks below threshold*			
Occupation overall exposure	**0.1482**		
Automation share = 0.097, augmentation share = 0.903 (Q4 overall)			

Task coverage: 7 of 20 tasks matched in the Claude conversation sample (35 %).

Worked example: many-to-one mapping



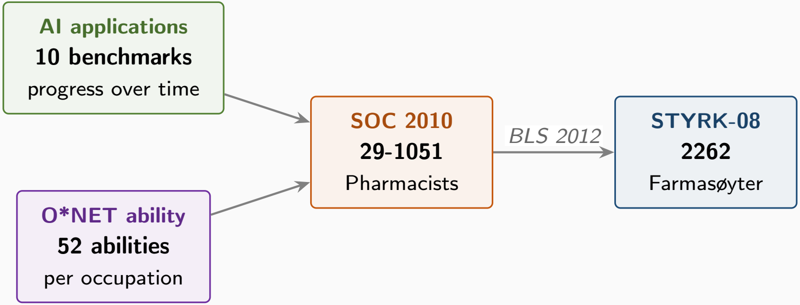
- Three SOC 2010 codes converge to STYRK 1112; share weights pool across all three
- 19 of 89 underlying tasks match a Claude conversation (21 % coverage)
- Automation share 0.532, augmentation share 0.468
- Generalist penalty: only 18.3 % of O*NET tasks appear in Claude conversations

Felten et al. (2021)

What Felten et al. measure

- Ten AI application benchmarks track progress over time:
  - image recognition, language modeling, speech recognition, reading comprehension, translation, abstract strategy games, real-time video games, visual Q&A, image generation, instrumental track recognition
- Each O*NET ability is rated for relevance to every AI application
- AIOE for an occupation = sum over abilities of (ability prevalence + importance) $\times$ AI progress on that ability
- Language Modeling sub-index restricts to the LM benchmark only

The code chain



Two inputs (AI progress, ability importance) combine at the SOC level. The downstream chain is identical to the other measures.

Worked example: Pharmacists (SOC 29-1051)

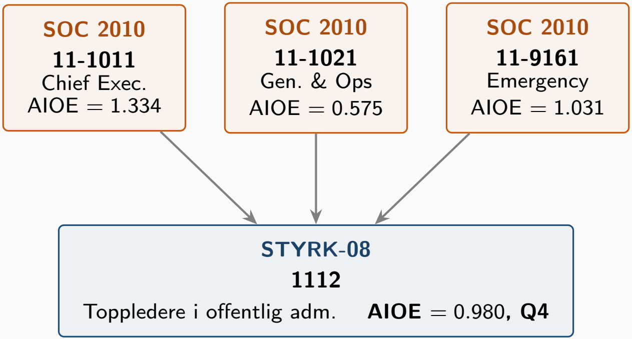
AI progress score for selected O*NET abilities (from Appendix D):

O*NET ability	Language Modeling	Reading Compr.	Image Recognition	Avg. across 10 apps
Deductive Reasoning	0.80	0.87	0.65	0.73
Inductive Reasoning	0.79	0.84	0.69	0.73
Written Comprehension	0.83	0.91	0.57	0.64
Oral Comprehension	0.91	0.69	0.37	0.61
Number Facility	0.50	0.64	0.67	0.63
Information Ordering	0.85	0.92	0.87	0.88
Memorization	0.86	0.93	0.83	0.84

AIOE weights these AI-progress scores by each ability's prevalence and importance in the occupation (from O*NET). For Pharmacists, the weighted sum gives:

AIOE $= 0.606$ **(Q4)** LM sub-index $= 0.571$ Image-Gen sub-index $= 0.478$

Many-to-one mapping



Unweighted mean across the three contributing SOC 2010 codes: $(1.334 + 0.575 + 1.031)/3 = 0.980$. The same averaging rule applies for Eloundou and Handa.

Anthropic (2026)

What the Anthropic 2026 `job_exposure` measure tracks

- Time-weighted Claude task coverage at the occupation level
- Augmentative use counts at half weight (automation and API use at full weight); tasks gated on Eloundou feasibility
- Released by Anthropic in March 2026 as an update to the Anthropic Economic Index
- Distinguishes itself from Handa by using time weights rather than conversation shares, and by re-weighting toward automation

The code chain

The public release reports a single `observed_exposure` score per occupation; the task-level decomposition is not exposed.

Worked example and cross-measure comparison

STYRK 2262, Farmasøyter (Pharmacists, SOC 29-1051)		
Anthropic `job_exposure`	0.090	Q4
Handa overall exposure	0.148	Q4
automation share	0.097	Q1
augmentation share	0.903	Q5
Eloundou β	0.429	Q4
Felten AIOE	0.606	Q4

Same occupation, same SOC code, same one-to-one mapping to STYRK 2262. Different scores reflect different underlying inputs: theoretical task exposure, ability-based AI progress, observed conversation shares, time-weighted usage. The quintile assignment is Q4 across all four measures here, but level differences would matter for any cardinal comparison.

Comparison

Four measures, one destination

Measure	Input unit	US code system	STYRK coverage
Eloundou (2024)	O*NET task (E_1, E_2)	SOC 2018	97.5 %
Handa (2025)	Claude conversation $\rightarrow$ O*NET task	SOC 2010	86.5 %
Felten (2021)	O*NET ability $\times$ AI progress	SOC 2010	95.8 %
Anthropic (2026)	Time-weighted Claude usage	SOC 2010	95.3 %

All four measures reach STYRK-08 via the BLS SOC$\rightarrow$ISCO crosswalk. They differ in the input unit (task vs. ability vs. conversation), in whether exposure is theoretical or revealed, and in whether automation and augmentation are separated.